Theory Notes
EconS 529 - Chouinard

Every paper has a theoretical foundation whether the author recognizes it or not. Be explicit about the theory driving your work.

Be sure you have a contribution (not a toy model). It's helpful to have a realistic application. Remember to have humility. You probably haven't made a huge leap forward in the literature. Now is the time to look carefully in the literature to find connections with what you are doing.

Theoretical paper:
To provide a logical and deductive argument that begins with your research question, expresses your proposed hypotheses and leads the reader through the analysis to the analytical results. This section will provide your analytical work providing answers to the hypotheses.

Components of the model section

1. Describe any existing theories used as the basis of your argument. If you use a common theory – don't derive well known theories, focus on how the question is an application of the theory
2. Modify existing theory
    A) Restate existing theory so reader knows what occurs in earlier work. Don't assume the reader knows the previous work.
    B) Introduce changes
        Include assumptions explicitly
        Discuss manipulations – explain how you move between each step
    C) Describe results
3. Develop new theory if don't rely on existing theory
    A) Describe the feasible space of the model
    B) Explicitly explain all assumptions
    C) Provide agent objective or relationship
    D) Discuss manipulations
    E) Describe results
        a. Lemmas, propositions
        b. Use to identify equilibrium
        c. Provide comparative statics
        d. Analytical or simulation

As you modify or develop a theory you may want to explicitly identify and describe definitions, assumptions, lemmas, propositions and corollaries. You will also need to provide formal proofs of lemmas, propositions and corollaries (perhaps in appendices).
Lemmas – a helping theorem used in the propositions.
Proposition - Theorem that can be proven from some previously accepted statements.
Corollary – Theorem that follows directly from a proposition.

Empirical paper:
WSU Graduate Student - “I really don’t care about the theory, I just want to apply it.”

To provide a logical and deductive argument that begins with your research question and ends in your proposed hypotheses. This section should set up the proposed cause and effect you are studying. This may be titled the theory or model or methods section, but every paper must include some of this information. You may provide less detail in this section for an empirical paper as the main contribution may not be providing new theory. Every paper (theoretical or empirical) must be based on economic theory and that must be expressed.

Components of the model section

1. Describe the theory used to make your argument
    i. Common theory – don’t derive well known theories, focus on how the question is an application of the theory
    ii. Modify existing theory
        a. Restate theory so reader knows what occurs in earlier work. Don’t assume the reader knows the previous work.
        b. Introduce changes
2. Include assumptions explicitly
3. Discuss manipulations – explain how you move between each step
4. Explain how the theory is relevant – relates to question
5. Explain how it diagnoses the problem – leads to hypotheses. You should use economic theory to derive a model that results in an expression of your hypotheses.
6. Comparative statics if possible and appropriate
7. Set up empirical model error term
8. **Bridge** between theory and empirics

Hints for all Theory Sections
Thomson, W. “A Guide for the Young Economist,” MIT Press: Cambridge, MA 2001.:

- Understand the role of your model. It is a tool. You don’t have to defend it as a complete representation of the real world, but you need to be careful what you explain with it.
- Introduce the model by moving from infrastructure to superstructure. Describe each element before you bring them together.
- Redundancy is useful, but don’t overdo it.
- Don’t assume readers will understand even very simple things. If there is something simple at the heart of an argument explain it.
- Don’t assume readers’ familiarity with your terms and definitions.
- Make it clear when you are defining a new term.
- Give examples illustrating novel definitions such as objects that satisfy the definition, do not satisfy it, that satisfy it but almost do not, or that don not satisfy it but almost do.
- Separate formal definitions from your interpretations.
- In a definition, don’t collapse several statements into one.
- Indicate what a definition depends on.
- Present basic concepts in full generality.

- Don’t use different terms for the same concept.
- Name concepts carefully.
- Avoid technical jargon.
- Use technical terms correctly.
- Use one enumeration for each type of object (definitions, assumptions, etc.).
- State assumptions in order of decreasing plausibility of generality.
- Group assumptions in related groups.
- Describe logical relationships between assumptions or groups of assumptions.
- Verify there are objects satisfying all your assumptions.

How to write a proof:

Reading: Cheng, E. 2004. “How to write proofs: a quick guide.” Working paper. Department of Mathematics, University of Chicago.

Proof hints:

- Choose right mixture of mathematics and words. This is true for derivations or mathematical manipulations in theoretical or empirical papers.
- Divide proofs into clearly identified steps or cases.
- Provide all the conditions needed to reach a conclusion.
- Don’t leave too many steps to the reader. What seems obvious to you may not to a reader.
- Be clear when a proof ends.
- Explore all possible variants of your results. Can you say something else? Have you said too much?

Notation hints:

- **Choose easily recognizable notation**. It’s good if readers can guess what the notation means.
- **Choose mnemonic abbreviations** for assumptions and properties (or avoid abbreviations). Don’t use A1 for assumption one if you expect readers to know what you mean by A1 several pages after you introduce it.
- **All notation should be easy to pronounce and draw on board** (for when you present the work.)
- **Don’t introduce notation you use only a few times.** This just adds unneeded clutter.
- **Choose notation that yields uncluttered mathematical expressions.**
- **Use single characters**. Generally, use single characters to represent variables to avoid confusion with acronyms or with two other variables multiplied together.
- **Minimize decorations**. Don’t add decorators such as overbars and carets unnecessarily. When necessary, use them for typical meanings. Overbars usually represent means, averages, or constraint limits. Carets often represent estimates, etc.
- **Minimize subscripts and superscripts subject to clarity**. Reduce subscripts and superscripts as much as possible. If your derivation applies to one of many agents but you never refer to more than a representative agent, then do not use a subscript to represent a specific agent.

- **Use subscripts to reduce repetitive material**. For example, rather than saying "expected profit is $p(e_1 - s_1\alpha) + (1-p)s_2\alpha$ for large growers and $q(e_1 - s_1\beta) + (1-q)s_2\beta$ for small growers", say "expected profit is $p_i(e_{1i} - s_{1i}\alpha_i) + (1-p_i)s_{2i}\alpha_i$, i = 1,2, where i = 1 for large growers and i = 2 for small growers. Avoid long lists by replacing "s_{11}, s_{12}, s_{13}, s_{21}, s_{22}, and s_{23}" with "s_{ij}, i,j = 1,2,3".
- **Use notation typical of the literature and target journal**. Try to use the same characters typically used in the literature to increase referee comprehension and familiarity.
- **Uncommon symbols**. Don't use uncommon notation like "$\perp$" if you are going to use it only a few times. If you using it frequently in a less technical journal economizes on space significantly, then define it on first use.
- **Appropriate symbols**. Don't use a period between to symbols to represent multiplication as in *x.y*. Either use the equation editor to represent it as $x \cdot y$ or find the comparable symbol by selecting "Insert" and then "Symbol" to represent it is *x·y*. Don't use "x" to represent multiplication. Rather, use the equation editor to write $x \times y$. Don't use and asterisk "*" to represent multiplication unless you are reproducing computer code.
- **Define all notation without delay**. Define all notation in an equation not previously defined in the same sentence it is introduced. Define notation and terms in the same order they appear in the equation.
- **Defining notation with clarity**. Define the dimension of variables carefully. In technical journals you might use "$x \in \Re_+^2$" to define what is described as a "positive two-dimensional variable" or "$x \in \Re^1$" to define what is described as a "scalar real variable" in a less-technical journal. Some journals require all vector and matrix notation to be bold faced. Follow the convention of your target journal. If a variable is an indicator variable that can take only two possible values, then indicate the two possible values when it is defined
- **Group notation definitions**. Generally, defining notation at one place makes it easy for the referee to refresh his/her memory of the meaning of symbols. This makes reading easier for the referee and may avoid confusion. A simple list form as follows suffices:

  p is the output price
  q is the output quantity
  π is net profit.

  Define terms in a logical order such as alphabetic. Define each term completely. For example, if it has a t subscript to represent time, then add "at time t".
- **Consistency of notation**. Once you give a symbol a name, stick with it. Don't call π "profit" at one place and "returns" at another. You will cause the referee confusion. Don't call something by interchangeable names such as "reservation utility", "reservation wage", and "reservation profit." Choose the best terminology and stick with it. If you redefine a term with each of several cases, you should probably add a subscript to represent the case. Failure to do so can lead to false conclusions when using the term more generally in comparing cases.
- **Use the proper decoration**. Don't use an accent mark such as □□ or apostro when you should use a prime. The character should appear exactly as the prime mark in the equation editor. In Microsoft Word, you can click on "Insert", then "Symbol", and choose "General Punctuation" to find ′ or ″.

- **Sophistication**. Economize on math and use mathematical sophistication appropriate to the target journal.
- **Punctuation of math**. All equations should be part of a sentence, whether displayed or not. Your mathematical expressions should have periods after them if at the end of a sentence or commas between expressions just as if they read as English.
- **Punctuation following mathematical terms**. You can avoid odd looking spaces between formulas produced by an equation editor and a subsequent punctuation mark. Put the punctuation mark in the end of the formula in the equation editor.
- **Avoid forward referencing**. Never reference an equation ahead of where you are in the manuscript.
- **Don't begin a sentence with notation**. Generally, do not begin a sentence with notation because the reader is not as well notified of where sentences begin and end. Make reading as easy as possible.
- **Use of equation numbering**. Use equation numbers liberally to remove ambiguity in referring to specific equations, results, or models. But do not number equations that are not referenced. Make sure all equation numbers are consecutive. Omitting an equation number to avoid renumbering is a sure sign of laziness and sloppiness.
- **Location of equation numbers**. Include equation numbers on the left or right margins according to instructions of your target journal. If equation numbers are on the right margin, set a right tab exactly at the margin so all equation numbers appear exactly at the margin, not simply randomly spaced to the right of the equation.
- **Mathematical assumptions**. Don't make assumptions simply out of convenience. Always justify assumptions intuitively and qualify them appropriately when they are introduced or the referee will reject your paper for making unjustified assumptions. Justification can be very brief for standard assumptions, e.g., "$u' > 0$ and $u'' < 0$ are assumed as is common." In some cases you can justify an obviously unrealistic assumption by arguing that the other generalities in your model are more important to that this simplification enables a meaningful abstraction of reality from which insightful results are possible.
- **Avoid undefined results**. Be sure that you aren't basing results on fractions where the denominator can be zero.
- **Beware of inconsistent assumptions**. Always make sure your assumptions are mutually consistent. This problem typically arises in trying to find qualitative comparative static results where results depend on the sign of complicated expressions. Consider every result you have from first- and second-order conditions and other assumptions before you simply assume a sign for a complicated expression or even state that some result occurs if some complicated expression has a particular sign. Through more careful mathematical manipulation a referee may find that an assumption or condition stated for convenience to establish the sign of another expression (or to satisfy a second-order condition) cannot hold in your model. This leads to sure rejection.
- **Strive for elegance**. Try to make your mathematics look as elegant as you can. Long messy formulas are foreboding to both referees and readers. More work is required to capture understanding. You goal is to convey understanding, not to look sophisticated. If you have expressions that are used repeatedly, try to define a single symbol to represent each. Try to give each a single verbal name that conveys intuition if you can. You will find that your own intuitive understanding as well as that of readers will increase.
- **Tell the story**. When you derive complex mathematical results, try to give an intuitive

explanation.

The more you can make your manuscript's appearance match articles in the published journal (aside from line spacing), the more seriously it will be considered. Polishing a paper like this may seem superficial but it can make a big difference in whether the substance of your paper gets properly considered.

- **Consistent font face and size**. All notation should be in the same font face and size (12 point) as the text in the paper. Make sure your equation editor does not modify notation size. A manuscript looks sloppy when the type size of the text is 12 point and perhaps mentions variables like Y_i and x_{2i}, and then an equation appears as

  $$Y_i = f(a_{1i} + x_{2i}) + \varepsilon_i$$

  or worse, in a different type face as

  $$Y_i = f(a_{1i} + x_{2i}) + \varepsilon_i.$$

- **Italics and bold facing**. If you italicize or bold face notation then it should be italicized or bold faced consistently, not just when it appears in the equation editor. While you can turn off italicizing notation in the equation editor, your paper will appear more professional if it is italicized consistently.
- **Spacing**. In Microsoft Word, spacing around expressions entered by the equation editor in running sentences can be awkward. Normally, if you insert a term or equation from an equation editor in a running sentence, add a space before and after the term or equation. Individual symbols and simple equations in running sentences often appears more professional when entered as running text using the "Symbol" table available under "Insert". But make sure you don't use φ in some places and ϕ in others. Both represent the same Greek character "phi."
- **Equations as running text**. You can often write equations as running text using the symbol table available in Mircosoft Word by clicking on "Insert" and "Symbol". But if you do, make sure your equations appear exactly as they do with an equation editor. This means place a space before and after every "=", "+", etc. For example, a paper looks less professional to have an equation produced by an equation editor such as

  $$Y_i = f(a_{1i} + x_{2i}) + \varepsilon_i$$

  followed by another that appears as $Y_i{=}f(a_{1i}{+}x_{2i}){+}\varepsilon_i$, or worse, as $\mathrm{Y_i{=}f(a_{1i}{+}x_{2i}){+}}\varepsilon_\mathrm{i}$. Without using the equation editor you can enter this as $Y_i = f(a_{1i} + x_{2i}) + \varepsilon_i$ so that it appears in a consistent form much as typeset material in a finished journal publication.

  - **Do not italicize parentheses, brackets, or braces**. A paper looks less professional if the above displayed equation is followed by $Y_i = f\mathit{(}a_{1i} + x_{2i}\mathit{)} + \varepsilon_i$
  - **Do not italicize punctuation, mathematical symbols, and numerals**. Do not italicize numerals, but italicize alphabetic characters if that is the way your equation editor does it.
  - **Italicizing Greek characters**. Italicize Greek characters or not just as your equation

editor does. For example, lower case Greek characters are often italicized but upper case Greek characters often are not.

Follow these rules even within subscripts or superscripts. When notation varies in sloppy and lazy ways, the referee will take less care in reviewing your paper. So if some notation is entered outside of the equation editor, it should appear exactly as if it were in the equation editor.

- **Consistent formatting**. When formats switch back and forth in your manuscript, it looks haphazard and conveys the impression that the research may also have been haphazard.
- **Fractions**. Representing fractions vertically in displayed equations usually makes reading easier, e.g.,

$$\frac{\partial f(x)}{\partial x} = \frac{wx}{\pi}(ax + bx^2),$$

  but vertical fractions look awkward in running sentences. If you put such an equation in a running sentence, then write it as $\partial f(x)/\partial x = (wx/\pi)(ax + bx^2)$.

- **Small fractions**. Small vertical fractions like ⅓ or $\frac{1}{2}$ are not commonly used.
- **Large symbols**. Similarly, when you use summations or integrals it is easier to read vertical representations such as

$$Z = \sum_{i=1}^{n} f(x_i)$$

  in a displayed equation but a subscript/superscript form such as $Z = \sum\nolimits_{i=1}^{n} f(x_i)$ makes the use of a summation appear much less awkward and disrupts the spacing much less in a running sentence. But, above all, follow the convention in your target journal.

- **Max, min, and lim operators**. When you have equations that involve maximization, minimization, limits, etc., always use the same form. That is, don't use "max" in one equation, "*Max*" in another, and "*maximize*" in another. It is best the use what the equation editor recognizes. Use

$$\max_{x,y} f(x, y \mid a)$$

  in a displayed equation, but using the form $\max_{x,y} f(x, y \mid a)$ in a running sentence makes the spacing appear more like a published article.

- **Clarifying the choice variables**. When you have equations that involve maximization, minimization, limits, etc., always indicate explicitly the variables to which they pertain. Don't leave the referee confused or free to guess.
- **Indicating conditions**. Usually, using a vertical bar rather than a colon to separate conditioning variables in a function or in set notation works better because a referee is less likely to miss it.
- **Nested parentheses, brackets and braces**. For equations that require multiple sets of parentheses, using brackets ("[" and "]") in place of outer sets of parentheses, or even

braces (“{“ and “}”) for extreme outset sets of parentheses, can make reading easier.

- **Use parentheses to avoid ambiguity**. Use parentheses to remove ambiguity in mathematical expressions. For example, does $1/2ce^2$ mean $(1/2)ce^2$ or $1/(2ce^2)$? In this case, the expression can be written unambiguously as $ce^2/2$ if it is not followed by another additive term. But generally a set of parentheses should be used around multiplicative terms intended to be in the denominator unless the fraction is written in vertical form.
- **Check for unbalanced parentheses**. Always double check to make sure your parentheses in an equation (or sentence) are balanced.
- **Avoid unnecessary parentheses**. Don’t use parentheses when they are not necessary. They only add to the reader’s burden. For example, if a sentence includes the term $x + y$, it need not have parentheses around it.
- **Long equations**. Any equation that is almost a line in length should be displayed. If you have equations more than a line in length, don’t permit wrapping within parentheses, brackets, or braces as in

$$
\begin{aligned}
CE = E[&t_1 + \delta t_2 + s_1\pi_1 + \delta s_2\pi_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2] \\
&- \tfrac{1}{2}\eta \,\mathrm{var}\begin{bmatrix} t_1 + \delta t_2 + s_1\pi_1 + \delta s_2\pi_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2)t_1 + \delta t_2 + s_1 e_1 \\ +\delta s_2 e_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2 - \tfrac{1}{2}\eta s_1^2(\sigma_1^2 + \sigma_\alpha^2) + \delta^2 s_2^2(\sigma_2^2 + \sigma_\alpha^2) \end{bmatrix}.
\end{aligned}
$$

  Rather, break the equation into chunks that fit within individual lines and add parentheses, brackets, or braces accordingly as in

$$
\begin{aligned}
CE = E[&t_1 + \delta t_2 + s_1\pi_1 + \delta s_2\pi_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2] \\
&- \tfrac{1}{2}\eta \,\mathrm{var}[t_1 + \delta t_2 + s_1\pi_1 + \delta s_2\pi_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2)t_1 + \delta t_2 + s_1 e_1 \\
&+ \delta s_2 e_2 - \tfrac{1}{2}ce_1^2 - \tfrac{1}{2}\delta ce_2^2 - \tfrac{1}{2}\eta s_1^2(\sigma_1^2 + \sigma_\alpha^2) + \delta^2 s_2^2(\sigma_2^2 + \sigma_\alpha^2)].
\end{aligned}
$$

  Try to break the equation at meaningful points that will aid readability for the referee. Also, indent the continuation of an equation consistently so extra lines are not confused. In MathType, you can click on “Style” and change to “Text” to add spaces at the front of a line for this purpose. Make sure line spacing within the formula matches the text, i.e., double spacing usually requires changing the line spacing. In MathType, you can change the spacing in the formula by clicking on “Format” and then “Line Spacing.” To get punctuation on the last line of the formula where it belongs, it must be added in the equation editor rather than after a multi-line equation produced by the equation editor.

- **Indenting equations**. Displayed equations are usually easier to pick out and read if they are indented compared to the running text, but indentation should be consistent. But above all, follow the format of the target journal.
- **Find elegant ways to express the mathematical material**.